TOPIC E: SME FINANCE COPILOT

Triody AI

A finance copilot. It warns a small business owner before cash runs out, finds a way through and never moves their money.

AL SIFAT SIDDIKI, Md Rezuan Islami Rokib, Atif Azeem
Edinburgh Napier University

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THE PROBLEM

A café owner sees the cash gap the day the bill lands

£9,200

Cash in the bank today. It looks fine.

£6,800

VAT due to HMRC on 7 November, as summer sales fade.

£1,500

Insurance renewal on 3 December, on top of rising wages.

A bank balance shows today. It says nothing about next month. The owner needs a warning weeks ahead, the reason behind it, and a way through.

WHAT TRIODY DOES

Four jobs, from the shop’s own books

Warn early

A 60-day cash forecast with ranges, and an alert before the balance drops under the owner’s buffer.

Explain why

Names the bills and cost trends behind the gap, in plain words.

Answer questions

“Profit in Q3?”, “What if sales drop 10%?”, typos and all.

Find financing

Ranks UK and China options, tests the best one on the forecast, drafts the paperwork.

DEMO: THE DAILY CHECK

Triody runs its own check and raises a critical alert

66%

chance the balance goes below zero in the next 60 days

−£1,194

lowest expected balance, on 7 December

24 days

of warning before the risk of breaking the £2,500 buffer reaches 25%

Cause, in Triody's words: the £6,800 VAT bill, the £1,500 insurance renewal, and supplier bills at 35% of sales with wages at 38%.

DEMO: FINANCING AND THE PLAN TEST

Best fit: pay £5,000 of the VAT bill over 10 months

HMRC Time to Pay, score 93/100

- Not a loan. No money is paid to the café.
- About £179 interest, at 7.75% a year
- 10 payments of about £518, 47% of free cash flow (limit 50%)
- Next: overdraft (74), iwoca (61). 6 products ruled out, each with a reason.

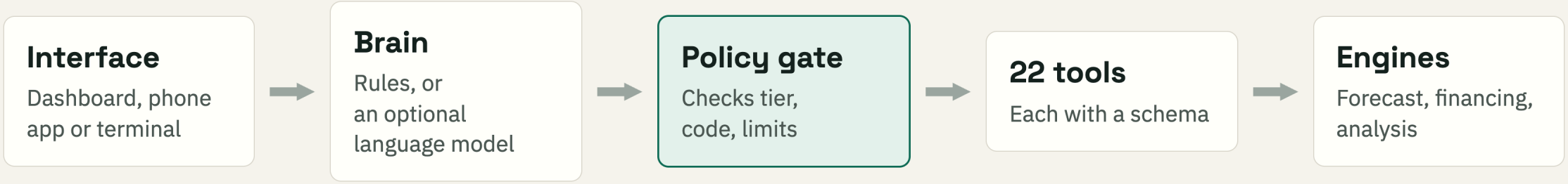
Tested on the forecast

Chance of going below zero	Without plan	With plan
Next 60 days	66%	13%
Whole plan, to Sept 2027	73%	59%

Triody tells the owner: the plan buys time, but it does not close the gap on its own.

ARCHITECTURE

Every request takes one path through one gate



Audit log
Every step, hash-chained with SHA-256

Owner device
Gets approval codes, never the chat

Runs anywhere
Python standard library only, offline in a browser via Pyodide

CORE ALGORITHM

The forecast gives a range, and sets its own width

1. Sales

Weekday pattern from the last 28 days. A damped trend. Big one-off orders capped at 1.5× the median, so one wedding order does not lift the forecast.

2. Bills

Finds weekly, two-weekly and monthly payments in the ledger. Supplier bills follow last week’s sales. Adds bills the owner lists, such as VAT.

3. 2,000 futures

Monte Carlo over 60 days gives p10, p50, p90 for every day. Range width comes from the shop’s own past 14-day errors.

Alert rule: warning when the chance of breaking the buffer reaches 25%. Critical when the expected path goes below zero, or half the futures do.

SECURITY: PERMISSION TIERS

Code decides what the agent does, not the model

Tier	Tools	Rule enforced in code
READ	17: forecast, reports, matching	Always allowed, arguments checked
PROPOSE	2: draft application or request	Allowed, nothing leaves the sandbox
EXECUTE	2: submit or send a draft	Owner only, with a one-time code
PROHIBITED	1: transfer_funds	Refused for everyone, even with a code

> A language model sees only READ and PROPOSE tools. Two tests check the payload sent to it.

> Memo text stays hidden from the model, which blocks prompt injection through a supplier memo.

SECURITY: OWNER APPROVAL

Nothing gets sent until the owner types the code

1

Triody drafts the application. A random 48-bit code gets bound to the exact action.

2

The code goes to the owner device only. Never the chat, the model or the log.

3

The owner types it. Limits checked: £50,000 per action, 3 actions a day.

4

The action runs once. The code expires after 10 minutes either way.

Five wrong codes lock every waiting draft. Every step lands in a SHA-256 hash-chained audit log, and the demo log verifies on all 41 entries.

EVIDENCE

Tested against simple baselines, labels frozen first

Test	Triody	Baseline
Financing match, 30 labelled cases	86.7%, 6/6 refusals right	76.7%, 4/6
Financing match, 14 held-out cases	85.7%, 3/3 refusals right	71.4%, 1/3
Financing match, 15 new held-out cases	93.3%, 2/2 refusals right	86.7%, 1/2
Day-14 balance error, café / bike shop	£672 / £1,434	£1,441 / £2,721
Day-14 balance error, florist / tea shop	£524 / ¥8,060	£793 / ¥25,793

192 unit, integration and safety tests, all passing

4/4 shops where the balance forecast beats the naive one

Baseline: cheapest eligible product for financing, last 28 days carried forward for balance. docs/EVALUATION.md. | 10

KNOWN LIMITS

What we have not proved yet

Limit	What a real deployment needs
Synthetic data only, four fictional shops	A pilot on real books, 12 months of history
Tea shop weekly sales: 7.12% error, baseline 6.61%	More seasonal data before trusting week-level sales
Language-model path tested with scripted models only	Run llm-check with six attack prompts on a live model
Owner device is a panel on the same screen	SMS or a push message in the bank’s app
Catalogue rates go out of date	Live quotes from providers, legal review

All 26 known risks, each with likelihood, impact and a fix, sit in docs/SECURITY_ASSESSMENT.md.

TRIODY AI

Warns early. Explains why. Prepares the fix. Never moves the money.

TRY IT

Python 3.11, no extra packages. Run run_mac.command or run_windows.bat, or open the browser app offline.

TEAM

AL SIFAT SIDDIKI (lead), Md Rezuan Islami Rokib, Atif Azeem.
Edinburgh Napier University.